Enhancing Delay Propagation Modeling Using Environmental and Operational Data

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Write a better introductory statement

1 INTRODUCTION

Train delays tend to occur quite frequently in the Dutch railway system, using machine learning models, the aim of this project is to identify the causes and therefore help prevent train delays. Although there has been research.

1.1 Objective

Develop a comprehensive model to predict and mitigate delay propagation in the Dutch railway system by integrating environmental factors (e.g., weather) and operational data (e.g., maintenance schedules, occupancy levels) alongside network topology.

1.2 Research Motivation

Complex Delay Causes: While the previous thesis focused on topological features, delays often result from multiple interacting factors.
Real-World Relevance:
Integrating diverse data sources makes the model more robust and applicable to real-world railway operations.

1.3 Research Question

How can environmental and operational factors be used to improve the prediction of delay propagation on the Dutch railway system?

- How do environmental factors (e.g., weather, temperature) influence delay propagation in the Dutch railway system?
- What role do operational factors (e.g., maintenance schedules, staffing levels) play in exacerbating or mitigating delays?
- How can machine learning models incorporating these factors improve on topology-only models in predicting delay propagation?

2 RELATED WORK

2.1 Foundation

This research builds on the foundation of the research conducted on US domestic airways by Lei et al. (2022) [2] and the following forecasting studies conducted on the Dutch railway network by Wen et al. (2018) and Li et al (2018) [6] [3]. This research has been inspired by a previous thesis, which forecasted delay propagation on trajectories solely on topological features.

2.2 Helpful sources

De Weert et al. (2024) is focused on maintenance scheduling using a Mixed Integer Linear Program to minimize train delays. This paper is unrelated to forecasting, but it provides insight on the maintenance aspect of this thesis. [1] Spanninger et al. wrote a paper comparing different train delay prediction approaches. This finds data-driven approaches to be more accurate overall; however event driven approaches lead to more explainable predictions. This paper could provide some extra justification for the approach my thesis will be taking.[5] Zhang, Liao et al (2021) propose using a Gradient boosting decision tree algorithm for train delay prediction. This algorithm functions somewhat similarly to xgboost and could provide inspiration for the methodology. [7]
Markovic et al. (2015) decribes an approach using support vector regression, this could be an interesting alternative to xgboost if implementation does not pose too many technical difficulties. This paper is also interesting as the reseach is conducted on the British railway system. The UK has a similar climate to the Netherlands. However this article is nearly 10 years old. [4]

3 METHODOLOGY

3.1 Data Collection

Firstly, the necessary data needs to be gathered and preprocessed. NS has made delay data on the Dutch railway network publicly available, this data includes the delay time and delay causes from 2019 until present, it also includes maintenance and other disruptions. Secondly the environmental features will be sourced from the Dutch weather dataset provided by KNMI, this includes weather data from different weather stations, for some stations even dating back to 1910. The weather at each train station for each day can thus be accurately derived from the KNMI dataset.

3.2 Exploratory Data Analysis

What trajectories are most affected by disruptions and what are the causes of these disruptions? What trajectories are most affected by weather? What trajectories are most often under maintenance/construction?

3.3 Feature Engineering

Combine environmental and operational data with network topology features (e.g., centrality measures).

3.4 Model Development

Train machine learning models (e.g., XGBoost, Random Forest) and evaluate performance with and without additional features. Alternative: Create a dynamic graph-based framework to identify and predict vulnerable routes within the Dutch railway network, enabling proactive measures to prevent cascading failures.

3.5 Validation

Multiple models will be compared across multiple delay scenarios, once with topological features and once with topological, environmental, and operational features.

4 RISK ASSESSMENT

4.1 Data

There do not seem to be any significant risks with the data as the data has already been used in previous research and is all publicly available.

4.2 Methodology

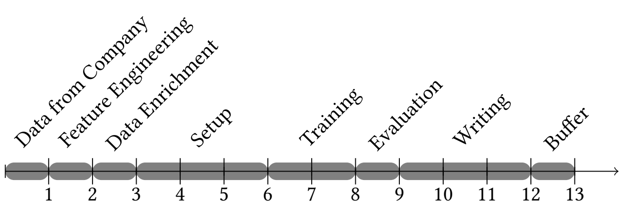
As the methodology is based on Lei et al., there should not be any major technical issues or challenges.[2]

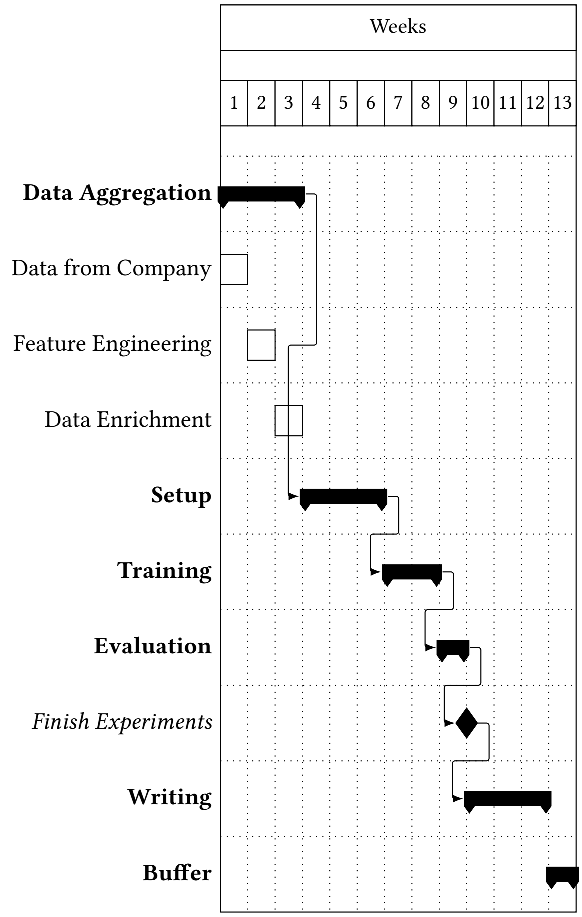
4.3 Results

If the results of using environmental and operational features do not seem to or negatively affect accuracy of the models, this project would still highlight disproportionately affected trajectories which is valuable information for improving the Dutch railway system.

5 PROJECT PLAN

Present your timeline here. Finally, you show your academic maturity by being able to quantify how much time your work will take (realistically!). Your UvA supervisor must be able to use your visual timeline to check whether you are on schedule. You can either use a timeline as below or a Gantt chart shown on the right.





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